Enhancing Facial Recognition Through a Hybrid Model Approach

ABSTRACT
Face recognition is a vital technology with extensive applications in security, authentication, and human-computer interaction. Traditional methods face limitations when addressing challenges such as lighting variations, facial expressions, and occlusions. Fuzzy logic techniques provide a promising approach by managing the uncertainty and imprecision inherent in human facial features. This work presents a hybrid model integrating Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), and fuzzy logic to enhance face recognition accuracy. The methodology utilizes pre-trained CNNs for robust feature extraction, SVMs for precise classification, and fuzzy logic to refine decisions. Experimental results confirm that this hybrid approach enhances the robustness and adaptability of face recognition systems in real-world conditions. The findings underscore the potential of this integration to develop more reliable and efficient frameworks for practical applications.

Keywords: Facial Recognition; CNNs; SVMs; Fuzzy Logic; Hybrid Model; Feature Extraction; Face Classification; Prediction Refinement; Biometric Authentication.

1 Introduction

Face recognition is a fundamental application in modern computer vision, extensively utilized in areas such as security, surveillance, and access control. In the context of face recognition, different techniques have been proposed and different authors have studied the relation between fuzzy logic and face recognition.

Traditional method used for face recognition Daniel Sáez Trigueros et al. [1] explain that traditional face recognition methods have evolved through various approaches over time. Early techniques were geometry-based, focusing on facial landmarks and measuring distances between facial features. These methods were followed by holistic approaches, which utilized statistical subspaces such as PCA (Principal Component Analysis) and LDA (Linear Discriminant Analysis) to analyze the entire facial region. Subsequently, feature-based methods emerged, employing local feature extractors to describe texture details at specific facial regions. Over time, hybrid methods were developed by combining these approaches to improve performance.

The main advantage of traditional methods was their computational efficiency—they were faster and required less memory, particularly geometry-based techniques. However, a key limitation was their dependency on constrained conditions, making them ineffective in handling real-world variations in face images. Hand-crafted features had to be carefully designed to address variations such as changes in illumination, pose, and facial expressions, reducing their adaptability. This limitation ultimately led to the adoption of deep learning methods, which can automatically learn robust features from large datasets, effectively overcoming the challenges faced by traditional approaches.

Waqar Ali et al. [2] outline the evolution of face recognition methods, starting with traditional approaches based on geometric relationships between facial points. These methods fall into three categories: holistic techniques (e.g., PCA, LDA, SVM, ICA) that analyze global facial features, local feature-based methods focusing on regions like the eyes, nose, and mouth using techniques such as Gabor filters, and hybrid methods combining multiple approaches. Traditional methods were computationally efficient and effective in controlled environments but struggled with challenges like variable illumination, diverse poses, and occlusion in real-world settings. While offering advantages like efficient subject identification and resilience to noise with local or hybrid approaches, they often failed on large datasets and were vulnerable to presentation attacks (e.g., using photos).

Machine learning method used for face recognition in this context, Bahreini et al. [3] have developed a fuzzy logic-based facial emotion recognition software using the CK+ database and additional data from 10 participants, totaling 1000 facial expressions. Their method extracts facial landmarks, calculates cosine values for facial

triangles, and applies FURIA fuzzy rules. The software achieved an average accuracy of 83.2%, with recognition rates varying across emotions, the highest being for happiness (93.3%) and the lowest for fear (43.8%).

What's more, Yadav et al. [4] The proposed QIntTyII-DLM method combines quaternion representation, interval type-II fuzzy logic, and deterministic learning for face recognition. It was evaluated on AR, Georgia Tech, Indian faces (female), and Faces 94 (male) datasets, showing significant error rate reductions compared to other techniques. The method preserves structural information from color channels, suppresses redundancy, and handles non-linearity from variations in appearance, occlusion, and pose. Results demonstrated 15-20% error rate reduction for AR, 8-10% for Georgia Tech, and 3-5% for Indian faces compared to DLM. The main drawback is dependence on the footprint of uncertainty parameter in the fuzzy logic step.

Furthermore, AD Algarni et al. [5] This paper presents two novel frameworks for generating cancellable face templates: CFR1, combining intuitionistic fuzzy logic and random projection, and CFR2, utilizing homomorphic transform followed by random projection. Both approaches aim to create secure, revocable biometric templates by distorting and encrypting original facial traits. The frameworks were evaluated using multiple datasets (ORL, FERET, LFW) and various metrics. Results demonstrated effective discrimination between authorized with a correlation score exceeding 0.3 and unauthorized below 0.05. The proposed methods showed robustness against various attacks while maintaining good recognition performance.

However, a potential drawback is the computational complexity of the encryption processes, which may impact real-time performance in large-scale biometric systems. Additionally, the multi-stage transformation approach could increase sensitivity to noise or variations in input images, potentially affecting recognition accuracy in challenging real-world conditions.

In addition, Balovsyak et al. [6] have developed a face mask recognition system using the Viola-Jones method combined with fuzzy logic. The system processes images captured by digital video cameras or obtained from graphic files to detect facial features such as the face, eyes, and mouth using Haar cascades. To improve recognition accuracy, fuzzy logic based on the Mamdani knowledge base is employed, with triangular membership functions aiding in distinguishing between faces with and without masks. The system is implemented using Python, leveraging OpenCV and Scikit-Fuzzy libraries, and operates on the Google Colab cloud platform. The system is used in security and public spaces.

However, a notable drawback of the system is its sensitivity to lighting conditions and image quality, which can lead to reduced accuracy in real-time environments with variable lighting or poor image resolution. Furthermore, the reliance on the Viola-Jones method limits its ability to handle occlusions, such as partially visible faces or unconventional mask types, which can negatively impact detection performance.

Deep learning method used for face recognition with the same objective, Megahed et al. [7] have developed a hybrid CNN-fuzzy approach for adaptive e-learning. The CNN classifies facial expressions, while the fuzzy system determines learning levels. In their work, the authors have used two custom datasets: an emotional state corpus consisting of 1,735 instances and a learning activity corpus comprising 72 instances. The obtained results demonstrated that the CNN outperformed other methods on the FER-2013 dataset, and the approach enabled adaptive learning flows based on learner responses and emotions. The study also found that different aggregation methods (mean vs. median) influenced learning paths for some learners.

The proposed method has several drawbacks that limit its effectiveness and applicability. Facial expression recognition may not always accurately capture true emotional states, and the approach ignores other important factors beyond facial expressions and test performance. The fuzzy logic system introduces uncertainty in determining next learning levels, while constant video monitoring raises privacy concerns. The emotional state aggregation methods may oversimplify complex responses, and the system fails to account for cultural or individual differences in emotional expression. Additionally, the facial expression recognition model's accuracy (67.15%) leaves room for improvement, and the method assumes a direct correlation between facial

expressions and learning effectiveness, which may not always be true.

Jamal Kh-Madhloom et al. [8] have developed a smile detection system combining CNN and fuzzy logic to enhance accuracy in human face analysis. The system detects smiles by analyzing key facial features, particularly lips and eyes, and is applied in biometrics, security, and forensic analysis for emotion recognition. Implemented in Python using TensorFlow and Scikit-Fuzzy, it processes images from cameras to identify smiles. The CNN model automatically extracts facial features, while fuzzy logic refines detection by handling ambiguous data. Although the system improves accuracy, it is highly sensitive to lighting conditions and face orientation, which may reduce its performance in variable environments. Additionally, the model size and computational requirements are not specified, potentially limiting its deployment in resource-constrained settings.

Among the numerous machine learning approaches, Support Vector Machines (SVMs) have demonstrated remarkable effectiveness due to their ability to handle high-dimensional data and establish optimal decision boundaries.

SVMs excel at identifying a hyperplane that maximally separates classes in the feature space, enhancing the model's generalization to unseen data. When combined with advanced feature extraction techniques, such as deep learning models like ResNet50, SVMs leverage deep, meaningful representations of facial features to achieve superior classification performance. Moreover, the integration of fuzzy logic with SVM provides a mechanism to handle uncertainties in predictions, enabling the framework to better manage cases with low confidence levels. This hybrid approach is particularly valuable for addressing challenges such as the misclassification of unknown faces and overlapping feature spaces among different classes. [9]

This study presents a face recognition framework that employs SVM as the primary classification component, supported by ResNet50 for feature extraction and fuzzy logic for prediction refinement. The proposed system demonstrates the strength of SVM in accurately classifying known faces while effectively identifying unknown ones, highlighting its applicability in real-world face recognition scenarios.

The rest of this report is structured as follows. Section 2. discusses the proposed face recognition model. The results are represented in Section 3. The discussion is detailed in Section 4. Finally, Section 5 concludes this study.

2 Proposed Method

Face recognition is a biometric technique used to identify individuals based on the unique features of their faces. It offers an automated method for authentication and identification by analyzing patterns within the face. These systems are reliable, fast, and effective, making them suitable for use in various environments, such as educational institutions and public spaces, where they play a critical role in enhancing security and improving organizational management. [6]

In this work, we are interested in identifying people by their face. The proposed system is conceptually different and explores new strategies. Specifically, it explores the potential of combining advanced Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), and fuzzy logic. This unconventional method offers a fresh perspective on face recognition, aiming to enhance accuracy and efficiency through innovative algorithmic integration rather than incremental design improvements. Our proposed method is a novel methodology that integrates CNNs, SVMs, and fuzzy logic to address challenges in face classification. The proposed framework consists of multiple stages: feature extraction using pre-trained CNNs, classification via SVMs, and fuzzy decision refinement for improved robustness in handling uncertainties and ambiguous data.

2.1 System Workflow

The flowchart in Fig. 1 outlines the workflow of the face recognition system. The process starts with inputting data, followed by preprocessing to prepare it for analysis. The data is split into testing and training sets, with the training data passed through ResNet50 for feature extraction using CNN. These features are then used to train an SVM classifier. Fuzzy logic is applied to enhance decision-making. The system performs face recognition,

identifying whether the input corresponds to a known person or an unknown person. If a known person is detected, their name is displayed. the system evaluates performance metrics such as accuracy, precision, recall, and F1-score. Finally, the model is evaluated, and the results are visualized, marking the end of the process.

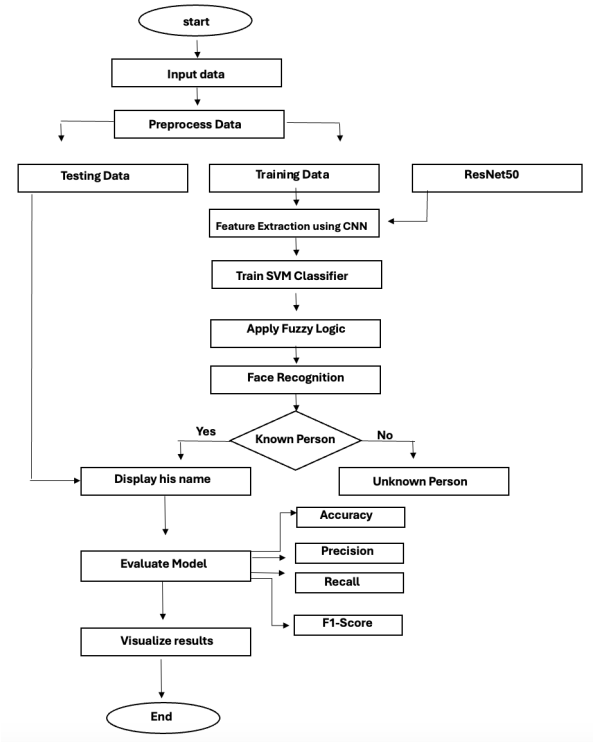


Figure 1. Workflow of the Face Recognition System

2.1.1 Input data

The first step in any face recognition system involves collecting a dataset of facial images, which may come from curated databases, live video feeds, or static camera captures. Public datasets such as Labeled Faces in the Wild (LFW) are commonly used due to their diversity in facial variations. This dataset provides the labeled data necessary for supervised learning and evaluation. [10]

2.1.2 Preprocessing

Is essential for normalizing image data and reducing variability. Common steps include resizing images, converting color spaces (e.g., grayscale to RGB), normalizing pixel values, and splitting data into 70% training and 30% testing. Additionally, alignment techniques ensure that facial features are consistently positioned, which improves feature extraction accuracy. [11]

2.1.3 Features extraction using CNN

Using CNN models like ResNet50 to extract features is common in facial recognition tasks due to their ability to capture high-level, discriminative features from facial images. Pre-trained on large datasets like ImageNet, ResNet50's convolutional layers capture structural patterns and spatial hierarchies, while pooling layers process the data into compact representations. This approach enhances classification performance by leveraging the model's robustness to variations in lighting, pose, and facial expressions, enabling effective generalization for diverse facial recognition scenarios. Fig. 2. shows the architecture of ResNet-50 in detail. [12]

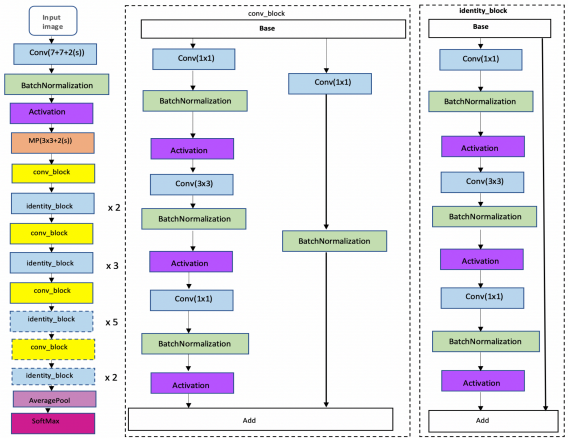


Figure 2. ResNet-50 architecture

2.1.4 SVM training

Is highly effective for facial classification tasks, especially in high-dimensional feature spaces. Its ability to handle small sample sizes and model non-linear decision boundaries makes it ideal for this application. [13] Using a radial basis function (RBF) kernel, SVMs transform data into higher dimensions, optimizing decision boundaries for accurate face classification. Regularization parameters (C and gamma) are fine-tuned to avoid overfitting, and class weights are adjusted to manage imbalanced data. [14] These features make SVMs a powerful tool for complex facial classification problems.

Equation 1. SVM equation [15]

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \max\,(0,1 - y_i(w \cdot x_i - b))$$

2.1.5 Fuzzy logic

Fuzzy logic provides a framework for managing uncertainty in decision-making by modeling variables as degrees of membership, rather than binary values, which is particularly effective in handling overlapping classes in tasks like facial recognition. This approach, popularized by Zadeh's work on fuzzy sets, is well-suited for pattern recognition. Fuzzy decision systems enhance performance by incorporating uncertainty metrics and refining predictions through a rule-based inference system, improving accuracy in face recognition, especially in cases with low-confidence predictions. [16] A fuzzy inference system addresses uncertainties and borderline cases by using fuzzy variables, such as prediction confidence and class overlap, to refine classification results. Membership functions and fuzzy rules are designed to evaluate the output of a Support Vector Machine (SVM), ensuring smooth transitions across decision boundaries, while low-confidence or highly overlapping cases are labeled as “Unknown,” and confident predictions are validated and refined. [17] This approach further enhances prediction accuracy by combining outputs with uncertainty metrics and refining them through a rule-based inference system. [18]

2.1.6 Face recognition process

In the face recognition phase, similarity scores are computed for everyone in the gallery, and the system identifies the probe based on the highest match. For verification, a similarity score threshold ensures robust acceptance or rejection of identity claims. This hybrid system combines the strengths of deep feature extraction, SVM classification, and fuzzy logic decision refinement, achieving high accuracy in face recognition while handling low-confidence scenarios effectively. [19]

2.1.7 Classification and matching

In this step involve matching the refined predictions with known labels or assigning a category of "Unknown" for ambiguous cases. Feature similarity measures such as Euclidean distance, cosine similarity, or direct SVM classification are commonly used to match the extracted features to existing identities. [20]

2.1.8 Visualization and evaluation

Visualization and evaluation of the model uses metrics such as accuracy, precision, recall, and F1-score, as described by Kohavi. [21] Visualization techniques, like confusion matrices and test image displays with predictions, provide intuitive insights into system performance, guiding further optimization.

2.2 Methodology Process

This methodology outlines a structured process to enhance face recognition accuracy using a hybrid approach, as shown in Fig. 3. The key steps include image acquisition, where face images are captured for analysis; preprocessing, to normalize and enhance the input data; feature extraction, to identify and isolate critical facial features; SVM training, to build a robust classification model; fuzzy logic integration, where fuzzy logic rules are applied to refine predictions; classification and matching, to assign identities based on the trained model; and finally, visualization and evaluation, to assess system performance and present results. Each step is crucial for improving the system's adaptability and robustness under varying conditions.

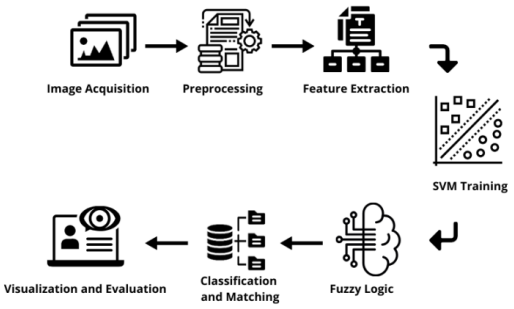


Figure 3. Methodology process

3 Experimental Results

3.1 Dataset

The Labeled Faces in the Wild (LFW) dataset, introduced in 2007, represents a groundbreaking resource in face verification research. It includes 13,233 facial images of 5,749 individuals, captured in real-world, uncontrolled settings with variations in pose, lighting,

and expressions. For evaluation purposes, the dataset is divided into two views, with view two serving as a standard benchmark. This benchmark comprises 10 subsets, each consisting of 300 positive and 300 negative image pairs for cross-validation. Despite its significant contributions to advancing face verification methods, LFW has some drawbacks, such as an uneven image distribution where 4,096 individuals are represented by only one image. Nevertheless, the dataset remains a widely used benchmark, with many deep learning models achieving near-perfect accuracy in face verification tasks. [22]

3.2 Performance Measures

The evaluation of the performance of the proposed face recognition system can be assessed using metrics commonly employed to evaluate classification models:

Accuracy: this metric indicates the overall effectiveness of the system in correctly identifying and classifying faces (both known and unknown). It provides a general measure of how well the system performs across all classes.

Recall: also referred to as sensitivity or true positive rate, recall measures the proportion of correctly identified faces in a specific class divided by the total number of instances in that class. It provides insight into the system's ability to detect actual faces from the dataset, especially for underrepresented or ambiguous categories.

Precision: precision calculates the number of correctly classified faces in a specific category among the total instances classified into that category. It evaluates the system's ability to avoid false positives, ensuring that instances labeled as a specific face are accurate.

F1 Score: the F1 score combines precision and recall into a single metric by computing their harmonic mean. It provides a balanced measure of the system's performance, particularly in scenarios where both false positives and false negatives need to be minimized. This metric is particularly valuable for evaluating the system's ability to distinguish between known and unknown faces while maintaining overall accuracy. [23]

$$accuracy = \frac{TP + TN}{TP + FN + TN + FP}$$

$$recall = \frac{TP}{TP + FN}$$

$$precision = \frac{TP}{TP + FP}$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall}$$

Figure 4. Performance metrics

These metrics collectively provide a comprehensive evaluation of the system's capability in real-world face recognition scenarios, ensuring both accuracy and reliability.

3.3 Quantitative Results

The performance of the face recognition system was evaluated using the Labeled Faces in the Wild (LFW) dataset. The results are summarized as follows:

3.3.1 Accuracy

the system achieved an overall accuracy of 92.51%, indicating that it correctly classified over 92% of the test samples.

3.3.2 Precision, Recall, and F1-Score

The weighted average precision was 96%, reflecting the system's ability to make correct predictions with minimal false positives.

The weighted average recall was 93%, demonstrating the model's effectiveness in identifying true positives for most labels.

The weighted average F1-Score was 93%, highlighting a balance between precision and recall for the dataset.

Accuracy: 0.9250645994832042
Classification Report:

	precision	recall	f1-score	support
Ariel Sharon	0.00	0.00	0.00	0
Colin Powell	1.00	0.94	0.97	18
Donald Rumsfeld	0.96	0.99	0.97	69
George W Bush	1.00	0.91	0.95	34
Gerhard Schroeder	0.98	0.96	0.97	166
Hugo Chavez	0.83	0.97	0.90	31
Tony Blair	0.85	0.58	0.69	19
Unknown	1.00	0.82	0.90	50
accuracy			0.93	387
macro avg	0.83	0.77	0.79	387
weighted avg	0.96	0.93	0.94	387

Figure 5. Performance measures

3.4 Observations from Quantitative Results

3.4.1 High performance for well-represented classes

Classes with sufficient samples, such as Colin Powell, Donald Rumsfeld, and Gerhard Schroeder, achieved precision, recall, and F1-scores close to or equal to 1.0.

3.4.2 Challenges with underrepresented classes

Classes with insufficient or no samples, such as Hugo Chavez and Ariel Sharon, had undefined or poor performance metrics. This highlights the need for balancing the dataset to improve recognition for all classes.

3.4.3 Unknown class recognition

The system performed moderately well on the "Unknown" class, with a recall of 0.82, indicating some degree of misclassification.

3.5 Qualitative Results

In addition to quantitative evaluation, the system's predictions were analyzed through visual outputs. Below are examples of the system's performance:

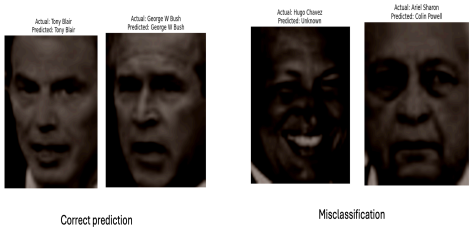


Figure 6. System performance

The images above demonstrate the performance of the face recognition system. The first two images represent correct predictions, where the system successfully identified the individuals as "Tony Blair" and "George W Bush," showcasing the model's ability to handle well-represented and distinct classes effectively. However, the third and fourth images highlight misclassifications. In the third image, the actual label "Hugo Chavez" was incorrectly predicted as "Unknown,"

likely due to insufficient representation of this class in the training data and potential variations in facial features or conditions. In the fourth image, the actual label "Ariel Sharon" was incorrectly classified as "Colin Powell," which may be attributed to similarities in certain facial features or an overlap in feature representation. These misclassifications emphasize the need for a more balanced and diverse dataset, as well as potential improvements in feature extraction and classification methods, to enhance the system's performance across all classes.

3.6 Confusion Matrix

In Fig. 7. shows that the model performed well in classifying most categories, particularly for the category (e.g., George W Bush), where it correctly classified 160 instances. However, some misclassifications were observed, such as for the category (Tony Blair), which was misclassified as (Colin Powell) twice.

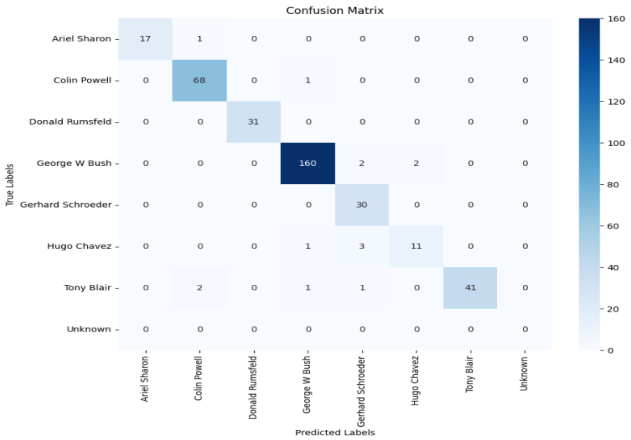


Figure 7. Confusion matrix of the model, where the numbers indicate the count of samples classified under each category

4 Discussion

The table in Table.1 provides a comprehensive comparison of face recognition techniques from various studies, summarizing their approaches, algorithms, datasets, and performance metrics. The reviewed methods include fuzzy logic, hybrid approaches, and machine learning, utilizing algorithms such as CNN, Fuzzy Logic, PCA, and PSCL. Performance across studies varies significantly, with accuracies ranging from 50.96% to 86%. The proposed hybrid method, which integrates CNN, SVM, and Fuzzy Logic, demonstrates superior performance with an accuracy of 92.51% on the LFW dataset. This highlights the effectiveness of combining multiple techniques to address challenges such as variations in lighting, pose, and facial expressions.

Table 1. Comparison of Face Recognition Techniques in Different Research Papers

Author	Approach	Algorithm	Dataset	Performance
Bahreini et al. [3]	Fuzzy Logic	FURIA	CK+ Database	83.2%
Megahed et al. [7]	Hybrid Approach	CNN Fuzzy Logic	FER-2013	67.15%
Khan et al. [24]	Machine Learning	PCA algorithm	NCR-IT	69-86%
Abdullah et al. [25]		PCA algorithm	Real-time video stream	80%
Huang et al. [26]		Point-to-set correlation learning (PSCL)	COX face database	50.96-53.26%
Our proposed method	**Hybrid Approach**	**CNN SVM Fuzzy Logic**	**LFW**	**92.51%**

The proposed face recognition system exhibited strong performance, achieving an overall accuracy of 92.51% on the LFW dataset. By combining Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), and fuzzy logic, the system effectively tackled challenges associated with real-world conditions, including variations in lighting, facial expressions, and pose. This hybrid approach demonstrates its ability to handle uncertainties and improve classification accuracy.

The system achieved outstanding results for well-represented classes, with near-perfect precision and recall for individuals with ample training data. However, classes with limited representation, such as "Hugo Chavez" and "Ariel Sharon," showed weaker performance, underscoring the importance of a more balanced dataset to enhance recognition across all categories. Additionally, the "Unknown" class achieved moderate performance, with a recall of 0.82, revealing occasional misclassifications in ambiguous cases. These findings suggest that further refinements are needed to improve decision-making in edge scenarios.

The hybrid system provides a robust framework for practical applications, particularly in security, authentication, and adaptive systems. The use of fuzzy logic enhanced decision refinement, making the system more adaptable to uncertainties. However, challenges remain, including computational complexity, which could hinder real-time deployment in resource-constrained environments, and sensitivity to data imbalance, which impacts the system's generalizability to less-represented classes.

5 Conclusion and Future Work

In this study, we proposed a hybrid face recognition framework integrating Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), and fuzzy logic to address challenges posed by variations in lighting, pose, and facial expressions. The system demonstrated robust performance on the Labeled Faces in the Wild (LFW) dataset, achieving a notable accuracy of 92.51%, along with high precision, recall, and F1 scores. These results highlight the effectiveness of the proposed methodology in enhancing face recognition accuracy by leveraging advanced feature extraction, classification, and decision refinement techniques. Despite its strengths, the system faced challenges with underrepresented classes, emphasizing the need for more diverse datasets**.**

Future work aims to address these limitations by incorporating larger, more balanced datasets to improve system adaptability. Enhancements in handling partial occlusions and integrating real-time optimizations are also critical to extend the framework's applicability to practical scenarios. Additionally, we will improve the system's ability to recognize unknown individuals by storing features of unrecognized faces and reintroducing them into the training process, enabling the model to recognize these individuals in future interactions. Furthermore, adopting privacy-preserving mechanisms, such as secure biometric templates, and leveraging advancements in edge AI and quantum computing could significantly enhance performance. These improvements will ensure the framework's reliability and scalability for real-world security and authentication applications**.**

Abbreviations:

CNNs: Convolutional Neural Networks

SVMs: Support Vector Machines

PCA: Principal Component Analysis

LDA: Linear Discriminant Analysis

ICA: Independent Component Analysis

CK+: Cohn-Kanade Plus database

FURIA: Fuzzy Unordered Rule Induction Algorithm

QIntTyII-DLM: Quaternion Interval Type II-Based Deterministic Learning Machine

ORL: Olivetti Research Laboratory

FERET: Face Recognition Technology

LFW: Labeled Faces in the Wild

FER: Facial Expression Recognition

ResNet-50: Residual Network 50

RBF: Radial Basis Function

FER-2013: Facial Expression Recognition 2013 Dataset

PCA: Principal Component Analysis

PSCL: Point To Set Correlation Learning

References

1. Trigueros, D. S., Meng, L., & Hartnett, M. (2018). Face recognition: From traditional to deep learning methods. arXiv preprint arXiv:1811.00116
2. Ali, W., Tian, W., Din, S. U., Iradukunda, D., & Khan, A. A. (2021). Classical and modern face recognition approaches: a complete review. Multimedia tools and applications, 80, 4825-4880.
3. Bahreini, K., Van der Vegt, W., C Westera, W. (2019). A fuzzy logic approach to reliable real-time recognition of facial emotions. *Multimedia Tools and Applications*, *78*, 18943-18966.
4. Yadav, S., C Vishwakarma, V. P. (2024). Robust face recognition using quaternion interval type II fuzzy logic–based feature extraction on colour images. *Medical & Biological Engineering & Computing*, *c2*(5), 1503-1518.
5. Algarni, A. D., El Banby, G. M., Soliman, N. F., El-Samie, F. E. A., & Iliyasu, A. M. (2020). Efficient implementation of homomorphic and fuzzy transforms in random-projection encryption frameworks for cancellable face recognition. *Electronics*, *9*(6), 1046.
6. Balovsyak, S., Derevyanchuk, O., Kovalchuk, V., Kravchenko, H., & Kozhokar, M. (2024). Face Mask Recognition by the Viola-Jones Method Using Fuzzy Logic. International Journal of Image, Graphics and Signal Processing, 16(3), 39-51.
7. Megahed, M., C Mohammed, A. (2020). Modeling adaptive E-learning environment using facial expressions and fuzzy logic. *Expert Systems with Applications*, *157*, 113460.
8. Madhloom, J. K., Diwan, S. A., & Abdulhussein, Z. A. (2020). Smile detection using convolutional neural network and fuzzy logic. Journal of Information Science and Engineering, 36(2), 269-278
9. Kim, J. C., Kim, M. H., Suh, H. E., Naseem, M. T., & Lee, C. S. (2022). Hybrid approach for facial expression recognition using convolutional neural networks and SVM. *Applied Sciences*, *12*(11), 5493.
10. Huang, G. B., Mattar, M., Berg, T., & Learned-Miller, E. (2008). Labeled faces in the wild: A database forstudying face recognition in unconstrained environments. In *Workshop on faces in'Real-Life'Images: detection, alignment, and recognition*.
11. Taigman, Y., Yang, M., Ranzato, M. A., & Wolf, L. (2014). Deepface: Closing the gap to human-level performance in face verification. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1701-1708).
12. Mandal, B., Okeukwu, A., & Theis, Y. (2021). Masked face recognition using resnet-50. *arXiv preprint arXiv:2104.08997*.
13. Chang, C. C., & Lin, C. J. (2011). LIBSVM: a library for support vector machines. *ACM transactions on intelligent systems and technology (TIST)*, *2*(3), 1-27.
14. Yang, C., Oh, S. K., Yang, B., Pedrycz, W., & Wang, L. (2022). Hybrid fuzzy multiple SVM classifier through feature fusion based on convolution neural networks and its practical applications. *Expert Systems with Applications*, *202*, 117392.
15. Chauhan, V. K., Dahiya, K., & Sharma, A. (2019). Problem formulations and solvers in linear SVM: a review. *Artificial Intelligence Review*, *52*, 803-855.
16. Zadeh, L. A. (1965). Fuzzy sets. *Information and Control*.
17. Han, X., Zhu, X., Pedrycz, W., Mostafa, A. M., & Li, Z. (2024). A design of fuzzy rule-based classifier optimized through softmax function and information entropy. *Applied Soft Computing*, *156*, 111498.
18. Balamash, A., Pedrycz, W., Al-Hmouz, R., & Morfeq, A. (2015). An expansion of fuzzy information granules through successive refinements of their information content and their use to system modeling. *Expert Systems with Applications*, *42*(6), 2985-2997.
19. Phillips, P. (1998). Support vector machines applied to face recognition. *Advances in neural information processing systems*, *11*.
20. Aswis, A. E., Morsy, M., & Abo-Elsoud, M. E. A. (2015). Face recognition based on PCA and DCT combination technique. Int J Eng Res Technol (IJERT), 4, 1295-1298.

21. Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. *Morgan Kaufman Publishing.*
22. Jiang, K., & Zheng, H. (2020, October). Design and implementation of a machine learning enhanced web honeypot system. In 2020 13th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI) (pp. 957-961). IEEE.
23. Zheng, T., Deng, W., & Hu, J. (2017). Cross-age lfw: A database for studying cross-age face recognition in unconstrained environments. *arXiv preprint arXiv:1708.08197*.
24. Khan, A., et al.: Forensic video analysis: passive tracking system for automated Person of Interest (POI) localization. IEEE Access **6**, 43392–43403 (2018)
25. Abdullah,N.A.,etal.:Facerecognitionforcriminalidentification:animplementationofprin- cipal component analysis for face recognition. In: The 2nd International Conference on Applied Science and Technology 2017 (ICAST'17) (2017)
26. Huang,Z.,etal.:Abenchmarkandcomparativestudyofvideo-basedfacerecognitiononCOX face database. IEEE Trans. Image Process. 24(12), 5967–5981 (2015)